COURSE DESCRIPTION			
Program	**Textile Technology**		
Course Title	**Technology of Manmade Fibres**		
Course Code	**2061**	**Semester**	**II**
Type of Course	**Theory**	**Contact Hours**	**60**
Teaching Scheme (L: T:P:R)	**4:0:0:0**	**Credits**	**4**
CIE Marks	**40**	**ESE Marks**	**60**

Introduction

This course provides a foundational understanding of the science and technology behind the production of regenerated and synthetic fibres. It covers polymer chemistry, spinning methods, fibre properties, and manufacturing processes of major regenerated and synthetic fibres. This course equips students with essential knowledge for careers in textile manufacturing, fibre development, and quality control.

Course Objectives

1	To provide fundamental knowledge on polymerisation concepts, fibre classification, and spinning methods used in man-made fibre production.
2	To impart an understanding of the manufacturing processes, essential properties and application of major regenerated and synthetic fibres.
3	To familiarise students with the features and applications of High-Performance fibres, and with the post-spinning processes that influence fibre performance.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Polymer chemistry	2003B	Chemistry for Technical Practices	I/II
Classification of textile fibres	1061	Fibre Science	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Classify the man-made fibres and understand the basics of polymerisation and spinning methods used in man-made fibre production.	Understand
CO2	Describe the manufacturing methods and properties of major Regenerated fibres.	Understand
CO3	Explain the production techniques, properties and applications of important man-made fibres.	Understand
CO4	Summarise the features and applications of important High-Performance and Speciality Man-made Fibres and outline key post-spinning processes.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	3	3	-	-	-	3	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-
Total	12	3	-	-	-	3	-	-	-	-	-
Correlation Level	3	3	-	-	-	3	-	-	-	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	

4	0	0	0	6	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Introduction to Man-made Fibres, Polymerisation, and Spinning Methods	Definition and Classification of manmade fibres. . Basic concepts of monomers, polymers and polymerisation. Different methods of spinning and the drawing process. Importance of staple fibres	15	0
Module II	Manufacture, Properties and Applications of Regenerated Fibres	Manufacture and Properties of Viscose Rayon. Basic Study of Acetate Rayon. . Brief Study of Cuprammonium Rayon. Overview of Modern Regenerated Fibres.	15	0
Module III	Manufacture, Properties, and Applications of Major Synthetic Fibres	Manufacture, Properties, and Uses of Polyamide Fibres (Nylon 6 and Nylon 6,6) Manufacture, Properties, and Uses of Polyester Fibres. Features and applications of other major synthetic fibres.	15	0
Module IV	Advanced Man-made Fibres and Post-Spinning Techniques	Features and applications of important High-Performance and Speciality Man-made Fibres. Outline key post-spinning processes of Man-made fibres.	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Classification of man-made fibres.	Define Regenerated and Synthetic fibres. List important regenerated fibres such as viscose rayon, acetate rayon, cuprammonium rayon, etc. Classify important synthetic fibres such as polyamides, polyesters, polyolefins, polyacrylonitrile, polyvinyl alcohol, polyurethane fibres, aramid fibres, etc., with suitable examples.	L	CO1	Understand	2
Basic concepts of monomers, polymers and polymerisation.	Define the terms monomers, polymers, polymerisation and degree of polymerisation.	L	CO1	Understand	2
Basic concepts of monomers, polymers and polymerisation.	Explain the classification of Polymers based on their structure(linear, branched and cross-linked) and intermolecular forces(Elastomers, fibres, thermoplastics and thermosetting polymers).	L	CO1	Understand	2
Basic concepts of monomers, polymers and polymerisation.	Describe the classification of polymerisation, such as condensation, addition and co-polymerisation	L	CO1	Understand	2
Different methods of spinning and the drawing process	Explain the different spinning methods—melt spinning, wet spinning, and dry spinning—with neat sketches.	L	CO1	Understand	3
Different methods of spinning and the drawing process	Explain the importance of drawing and differentiate between cold drawing and hot drawing.	L	CO1	Understand	2
Importance of staple fibres	State the difference between filament fibres and staple fibres. State the need for producing manmade staple fibres and their advantages.	L	CO1	Understand	2
Module II					
Manufacture and Properties of Viscose Rayon	Describe the manufacture of Viscose Rayon with a flow chart	L	CO2	Understand	4

Manufacture and Properties of Viscose Rayon	List the physical properties, chemical properties and uses of Viscose Rayon.	L	CO2	Understand	1
Manufacture and Properties of Viscose Rayon	State the salient features of Polynosic Fibres	L	CO2	Understand	1
Basic Study of Acetate Rayon.	Provide a brief description of the manufacture of Acetate rayon	L	CO2	Understand	2
Basic Study of Acetate Rayon.	List the physical properties, chemical properties and uses of Acetate Rayon.	L	CO2	Understand	1
Brief Study of Cuprammonium Rayon	Describe the key characteristics and uses of Cuprammonium Rayon	L	CO2	Understand	2
Overview of Modern Regenerated Fibres.	Explain the key features and uses of Lyocell, Modal, FR Viscose, and Bamboo fibres.	L	CO2	Understand	4
Module III					
Manufacture, Properties, and Uses of Polyamide Fibres (Nylon 6 and Nylon 6,6)	Explain the Manufacturing of Nylon 6 from Caprolactam and list the features of Nylon 6.	L	CO3	Understand	2
Manufacture, Properties, and Uses of Polyamide Fibres (Nylon 6 and Nylon 6,6)	List the physical properties, chemical properties and uses of Nylon 6	L	CO3	Understand	2
Manufacture, Properties, and Uses of Polyamide Fibres (Nylon 6 and Nylon 6,6)	Explain the features of Nylon 66 and list the physical properties, chemical properties and uses of Nylon 66	L	CO3	Understand	2
Manufacture, Properties, and Uses of Polyamide Fibres (Nylon 6 and Nylon 6,6)	Compare Nylon 6 and Nylon 66 in terms of the monomer used, type of polymerisation, chemical repeat unit, melting point, crystallinity, and strength.	L	CO3	Understand	1
Manufacture, Properties, and Uses of Polyester Fibres	Explain the Manufacturing of Polyester Filament (PET) from raw materials and list the features of Polyester filament.	L	CO3	Understand	2
Manufacture, Properties, and Uses of Polyester Fibres	List the physical properties, chemical properties and uses of Polyester.	L	CO3	Understand	1
Features and Applications of HDPE, LDPE, Polyolefin, Acrylic, and PVA Fibres	Differentiate between LDPE and HDPE. State features and uses of LDPE and HDPE.	L	CO3	Understand	1
Features and Applications of HDPE, LDPE, Polyolefin, Acrylic, and PVA Fibres	State the features and uses of Polyolefin fibres (Polyethylene and Polypropylene fibres) State the features and uses of Polyolefin fibres (Polyethylene and Polypropylene fibres)	L	CO3	Understand	2
Features and Applications of HDPE, LDPE, Polyolefin, Acrylic, and PVA Fibres	Distinguish between Acrylic and Modacrylic fibres. State the features and uses of Acrylic fibres.	L	CO3	Understand	1
Features and Applications of HDPE, LDPE, Polyolefin, Acrylic, and PVA Fibres	State the features and uses of PVA fibres.	L	CO3	Understand	1
Module IV					
Features and applications of various High-Performance Speciality Fibres	Explain the features and applications of important High-Performance Speciality Fibres such as Polyurethane (Spandex)fibres, Aramid (Nomex, Kevlar) fibres and Carbon fibres.	L	CO4	Understand	5
Post-spinning processes of Manmade fibres	Define texturisation and its objectives.	L	CO4	Understand	1
Post-spinning processes of Manmade fibres	Differentiate stretch yarn, modified stretch yarn and bulk yarn.	L	CO4	Understand	1
Post-spinning processes of Manmade fibres	Describe the principle and working of a false twist texturising machine.	L	CO4	Understand	2
Post-spinning processes of Manmade fibres	Explain the working of the Stuffer Box crimping machine.	L	CO4	Understand	1
Post-spinning processes of Manmade fibres	State the working of the Edge crimping machine.	L	CO4	Understand	1
Post-spinning processes of Manmade fibres	Explain the Air-Texturisation process and its salient features.	L	CO4	Understand	2

Post-spinning processes of Manmade fibres	Describe the knit-deknit texturising process.	L	CO4	Understand	1
Post-spinning processes of Manmade fibres	Explain the working of the gear crimping machine.	L	CO4	Understand	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Explore how linear, branched, and cross-linked polymers influence fibre flexibility and strength.	6
2	Study and explain why cellulose-based fibres cannot be melt-spun and require wet or dry spinning.	6
3	Explore how molecular weight and polymerisation degree affect the strength, elasticity, and durability of synthetic fibres.	6
4	Analyse why industries choose staple fibres over filaments for specific products like denim, blended yarns, or nonwovens.	6
Module II		
5	Identify the environmental concerns associated with viscose rayon production and list the modern solutions adopted by manufacturers.	6
6	Study the manufacturing process of Polynosic Rayon.	6
7	Study the manufacturing process of Cuprammonium Rayon.	6
8	Analyse why regenerated fibres are preferred in medical textiles, hygiene products, and apparel linings	6
Module III		
9	Study the importance of blending natural and synthetic fibres.	6
10	Explore the Manufacturing process of Nylon 6.6	6
11	Study the method of production of LDPE & HDPE.	6
12	Explore the method of production of Orlon fibres.	6
Module IV		
13	Prepare a brief report on how fibres like Kevlar, Nomex, Carbon, and Spandex are used in defence, firefighting, and industrial safety.	6
14	Study the method of manufacture of Polyurethane fibre.	6
15	Study smart fibres (antimicrobial, flame-retardant) and their relevance to modern textiles.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	A Textbook of Fibre Science and Technology	S P Mishra	New Age International Publishers
2	Handbook of Textile Fibres, II- Man-made Fibres	J Gordon cook	Woodhead Publishing Ltd
3	Man-made Fibres	Moncrieff R W	Wiley-Interscience publication

REFERENCE			
Sl. No.	Title	Author	Publication
1	Handbook of Textile Processors Series: The Substrates-Fibres, Yarn and Fabrics-	Mathews kolanjikombil	Woodhead Publishing India Pvt. Ltd
2	Yarn texturing technology	J W S Hearle, L Hollick and D K Wilson	Woodhead Publishing-Cambridge England

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	www.textilelearner.net	All

2	https://www.fibre2fashion.com/industry-article/texturizing	IV

Assessment Methodology – Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60